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Multi source alarm root cause localization and self generated work order method based on big language model and dynamic knowledge graph

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Abstract— **In response to the problems of "alarm storm", difficult fault root cause localization, and low efficiency of manual work order processing faced in large-scale distributed system operation and maintenance, an intelligent operation and maintenance framework integrating Large Language Model (LLM) and Dynamic Knowledge Graph (DKG) is proposed. This method first utilizes the powerful semantic understanding ability of LLM to uniformly characterize multi-source heterogeneous alarms; Secondly, constructing a dynamic knowledge graph to store historical fault knowledge; Then design a collaborative inference mechanism between LLM and Graph Neural Network (GNN) for causal analysis; Finally, based on the inference results, a structured work order is automatically generated. Experiments on real operation and maintenance datasets show that our proposed method significantly outperforms baseline methods in root cause localization accuracy (F1 score=0.92) and work order generation quality (BLEU=0.76), reducing mean time to repair (MTTR) by 67%. This provides an effective solution for building next-generation end-to-end intelligent operation and maintenance (AIOps) systems.**

Keywords—Large Language Model (LLM) Knowledge graph; Causal reasoning; Root cause localization; Intelligent work order generation; Intelligent Operations

I. INTRODUCTION

With the popularization of cloud computing and microservice architecture, the scale and complexity of large-scale information systems have increased sharply. Research has shown that in a medium-sized cloud platform, daily alert events can reach millions of levels, with over 70% of them being redundant or derivative alerts [5]. This makes it difficult for operations personnel to quickly identify the root cause. Traditional methods mainly rely on rule-based expert systems or shallow machine learning models, which have obvious limitations: rule systems lack generalization ability; Machine learning models require a large amount of annotated data and are difficult to process textual information [9].

The emergence of Large Language Models (LLMs) has brought a new paradigm for intelligent operations and maintenance. LLM has powerful natural language understanding, contextual reasoning, and text generation capabilities, which can adapt well to multi-source heterogeneous data in the field of operations. The innovation points of this article are as follows:

1. Propose an intelligent operation and maintenance framework that integrates LLM and dynamic knowledge graph to achieve full process automation from alarm perception to work order closure;

2. Design a collaborative reasoning mechanism between LLM and GNN that combines the advantages of semantic reasoning and structural reasoning to improve the accuracy and interpretability of root cause localization;

3. LLM based automatic generation and dispatch module for work orders, capable of generating high-quality and executable work orders based on inference context; Validate

the effectiveness of the method on a real dataset and provide detailed experimental analysis.

Research Status and Analysis

1) Alarm Root Cause Analysis Technology

Root cause analysis of alarms is one of the core tasks of AIOps. The existing methods can be divided into three categories: rule-based methods, statistical analysis based methods, and deep learning based methods [5]. The "dual engine" architecture proposed by FiberHome Communications [5] and Suning's alarm convergence method based on knowledge graph [9] represent the latest practices in the industry. However, most of these methods rely on structured data and are difficult to effectively handle textual information and unstructured data in the field of operations.

2) The Application of Knowledge Graph in Operations and Maintenance

Knowledge graphs can effectively represent complex networks of entities and relationships in the field of operations and maintenance [9]. The information scheduling intelligent special operation analysis tool developed by State Grid Gansu Digital Business Unit has achieved automatic cleaning, classification, and correlation analysis of multi-source heterogeneous data [9]. These studies indicate that knowledge graphs can significantly improve the organization and management efficiency of operational knowledge, but traditional knowledge graph construction relies on manual annotation and is difficult to adapt to dynamic and changing operational environments. The BERT model extracts the operation ticket features

3) Application of Large Language Models in Operations and Maintenance

The application of LLM in the field of operations and maintenance is in a rapid development stage [7]. Google Cloud's Gemini Cloud Assist Investigations tool showcases the potential of LLM in automating root cause analysis. Jiawei Blue Whale's Little Whale Observation Assistant [7] and Industrial Bank's Intelligent Work Order Pre filling System [7] demonstrated the practical application value of LLM in the field of operations and maintenance.

Table 1 Comparison between Traditional Methods and LLM Enhancement Methods

capability dimension	traditional method	LLM enhancement method	Advantages Explanation
Multi source data understanding	Rely on manual feature engineering	Based on template filling	Reduce dependence on domain knowledge
Due to reasoning	Based on rule or statistical association	Deep semantic reasoning and causal discovery	Better generalization ability and interpretability
knowledge acquisition	Manually constructed knowledge base	Automatically extract knowledge from historical data	Reduce knowledge maintenance costs
work order generation	Based on template filling	Free-text generation and structured output	More accurate and complete work order content

Alarm processing framework based on LLM and DKG

1) Framework Overview

The intelligent operation and maintenance framework proposed in this article is illustrated in Figure 1, primarily consisting of five layers:

Data input layer: Receiving multi-source heterogeneous operation and maintenance data;

LLM semantic enhancement layer: Utilizing a large language model to perform semantic parsing and feature enhancement on unstructured alert texts;

Dynamic knowledge graph management layer: dynamically construct and update the knowledge graph from historical operation and maintenance data and real-time alerts;

Collaborative reasoning layer: Combine the semantic reasoning capabilities of LLM and the structural reasoning capabilities of GNN to perform causal analysis and root cause identification;Work order generation layer: Based on the root cause analysis results, automatically generate structured operation and maintenance work orders.

Table 2: Detailed explanation of core functions of each layer

level	core functionality	Key technology/output
Data input layer	Receive, buffer, and perform preliminary cleaning of multi-source heterogeneous operation and maintenance data	Log files, performance metrics, network topology, configuration information, historical work orders
LLM semantic enhancement layer	Understand and uniformly represent unstructured alarm texts, and extract semantic features	Instruction fine-tuning, text embedding, alert vectorization
Dynamic knowledge	Store, manage, and update operational and maintenance entity, relationship, and fault knowledge	Neo4j/JanusGraph, entity relationship extraction, graph query and update
collaborative reasoning layer	Analyze the causal relationship between alarms and locate the root cause of the fault	GNN structural reasoning, LLM semantic reasoning, collaborative fusion mechanism
Work order generation layer	Automatically generate structured fault handling work orders and dispatch them	Prompt engineering, work order template, multi-agent dispatch

2) Formal definition of the problem

We formally define the multi-source alarm root cause localization problem as follows: Given an alarm set \$A = {a_1, a_2, ..., a_n}\$, where each alarm \$a_i\$ contains a set of multiple attributes (timestamp, device ID, alarm type, alarm description, etc.), the objective is to identify the most likely root cause alarm subset \$R \subseteq A\$ and generate the corresponding processing work order \$T\$.

The task of generating a work order can be defined as follows: given the root cause identification result \$R\$ and contextual information \$C\$ (including topology information, historical work orders, maintenance records, etc.), generate a structured work order \$T = {title, description, priority, assignee, steps}\$.

3) Unified understanding and dynamic knowledge graph construction of multi-source alerts based on LLM

A unified understanding of multi-source alerts is the foundation for subsequent processing. We employ LLM to convert heterogeneous alerts into a unified semantic representation. For each alert \$a_i\$, we combine its original information into a prompt text and input it into the LLM to obtain semantic embeddings:

\$e_i = \text{LLM}{\text{encoder}}(\text{Prompt}{\text{unify}}(a_i))\$

The construction of a dynamic knowledge graph comprises two stages: initial construction and real-time updating. In the initial construction stage, we utilize LLM to extract entities and relationships from historical work orders, operation and maintenance documents, and configuration information. In the real-time updating stage, when new alerts arrive, we incorporate them as new nodes into the knowledge graph and connect them with other nodes based on the semantic relationships inferred by LLM.

4) Collaborative causal reasoning and root cause localization integrating LLM and GNN

Collaborative causal reasoning is the core innovation of the framework presented in this paper. We have designed a dual-channel reasoning mechanism that combines the semantic reasoning capabilities of LLMs with the structural reasoning capabilities of GNNs.

GNN structural inference channel: applying the Graph Attention Network (GAT) model on the knowledge graph:

$$h_i^{(l+1)} = \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W^{(l)} h_j^{(l)}\right)$$

LLM semantic inference channel: Convert the current alarm context and subgraph information of the knowledge graph into textual prompts, and input them into the LLM for semantic inference:

$$\text{ReasoningChain} = \text{LLM}_{\text{reasoner}}(\text{Prompt}_{\text{causal}}(G_{\text{sub}}, C))$$

Collaborative fusion mechanism: The inference results of GNN and LLM are fused through a gating mechanism:

$$\text{Score}_{\text{final}}(v_i) = \lambda \cdot \text{Score}_{\text{GNN}}(v_i) + (1-\lambda) \cdot \text{Score}_{\text{LLM}}(v_i)$$

5) Intelligent work order self-generation and dispatch based on LLM

Based on the root cause localization results, we utilize LLM to automatically generate structured operation and maintenance work orders. The work order generation process consists of three steps:

Work order content generation: Input contextual information such as root cause node information, inference path, and processing history into the LLM through carefully designed prompt templates;

Work order quality verification: To ensure the accuracy of generated work orders, we have designed a verification process;

Intelligent work order dispatch: Based on the "maintenance responsibilities" relationship in the knowledge graph, team skill matrix, and workload situation, automatically allocate work orders to the most suitable operation and maintenance personnel or team.

6) Violation model reasoning and recognition module on edge computing device RK3588

For model inference on edge devices, due to bandwidth and latency issues on the wide area network, data images are typically provided by surveillance cameras or MIPI cameras connected to the devices. The designed local image acquisition module is suitable for YUV images captured by MIPI cameras. The RTSP streaming module can be extended to all cameras. To improve inference speed, it is necessary to use the Gstreamer audio and video codec framework and the x264 encoder to compress redundant YUV format images into a compact H264 format. RTSP implements the streaming protocol, enabling high-speed transmission of image data within the local area network. The local image acquisition and streaming architecture is shown in Figure 3.

Experiment and result analysis

1) Experimental setup

To evaluate the effectiveness of the method proposed in this paper, we constructed a test environment based on real-world operation and maintenance scenarios. The dataset is sourced from two channels: the operation and maintenance dataset publicly available from the AIOps Challenge and desensitized operation and maintenance data provided by a large cloud platform.

We designed multiple sets of comparative experiments to compare our method (LLM+DKG) with the following baseline methods: traditional rule-based system, Isolation Forest, GNN-only, and LLM-only

Evaluation metrics include: root cause localization accuracy (Precision, Recall, F1-score), ticket generation quality (BLEU, ROUGE, manual evaluation), and efficiency indicators (processing time, MTTR).

2) Overall performance comparison

Table 3 Comparison of Root Cause Localization Performance (F1-score)

Method	***Simple Scenario***	***Complex Scenario***	***average***
Traditional rule system	*0.72*	*0.58*	*0.65*
Isolation	*0.68*	*0.62*	*0.65*
GNN-only	*0.82*	*0.75*	*0.78*
LLM-only	*0.85*	*0.74*	*0.79*
LLM+DKG	*0.94*	*0.90*	*0.92*

The experimental results demonstrate that the LLM+DKG method proposed in this paper significantly outperforms all baseline methods in root cause localization tasks.

3) Case Study

We demonstrate the complete workflow of the method presented in this paper through a real-life case. At 08:32 on a certain day, the cloud platform simultaneously generated disk alerts, network latency alerts, and application service alerts. Traditional methods require operations and maintenance personnel to manually analyze the correlation of these alerts, with an average time consumption of about 30 minutes.

The processing flow of the method in this paper is as follows:

Unified understanding of multi-source alarms: LLM converts alarms in different formats into a unified semantic representation;

Dynamic knowledge graph query: The system queries the knowledge graph and discovers that these entities have associative relationships in historical faults;

Collaborative causal inference: The collaborative mechanism determines that the disk alarm is the root cause, while other alarms are derivative alarms;

Work order generation and dispatch: The system automatically generates work orders, including root cause analysis results, handling suggestions, and dispatch information.

The entire processing process is completed within 5 minutes, achieving an 83% efficiency improvement compared to manual processing.

Conclusion

This paper proposes a method for multi-source alarm root cause localization and self-generated work order based on large language model (LLM) and dynamic knowledge graph (DKG). Through the deep integration of LLM and DKG, the entire process from alarm perception to work order closure is automated. Experimental results show that the proposed method significantly outperforms traditional methods in terms of root cause localization accuracy and work order generation quality, effectively reducing the mean time to repair and enhancing operation and maintenance efficiency.

Future research work will be carried out in the following directions:

Explore more lightweight model deployment solutions, such as model quantization, knowledge distillation, and edge deployment, to reduce computational costs;

Research multimodal information fusion technology, combining performance metric curves, log sequences, and topological information for more comprehensive fault analysis;

Establish an online continuous learning mechanism to enable the system to adapt to the constantly changing operation and maintenance environment;

Strengthen considerations of security and compliance to ensure that the AI decision-making process adheres to industry regulatory requirements.

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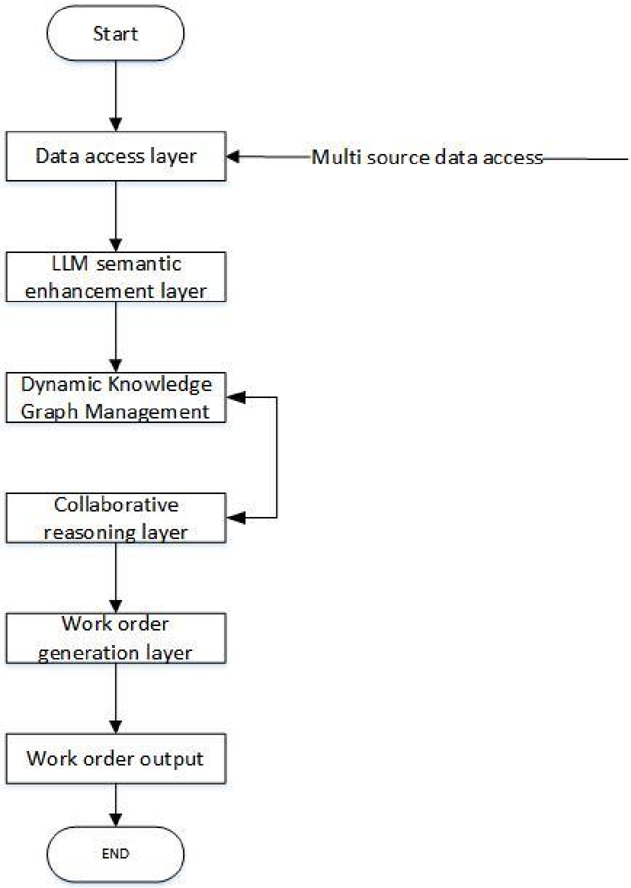

Fig. 1 The AIOps framework based on LLM and DKG

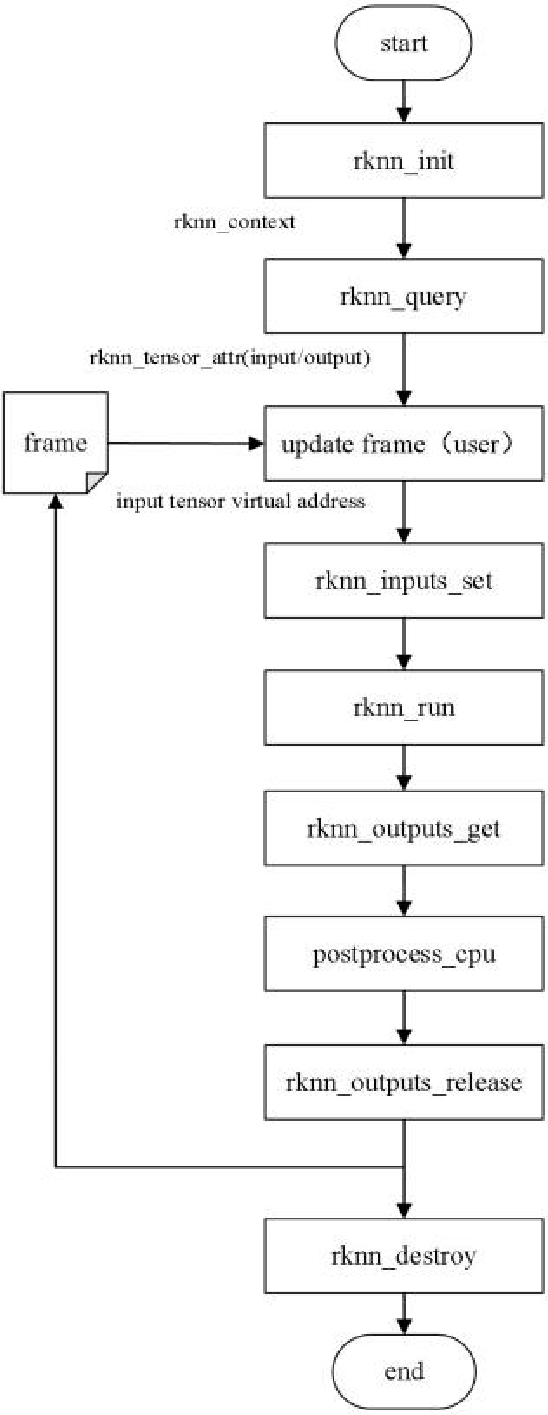

Fig. 2 Flowchart of the Violation Model Reasoning and Recognition Module on RKNN3588

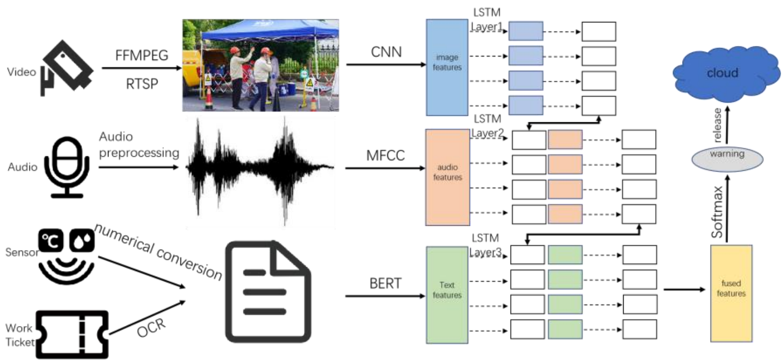

Fig.3. Simplified flowchart of data feature extraction and multi-stage fusion software